

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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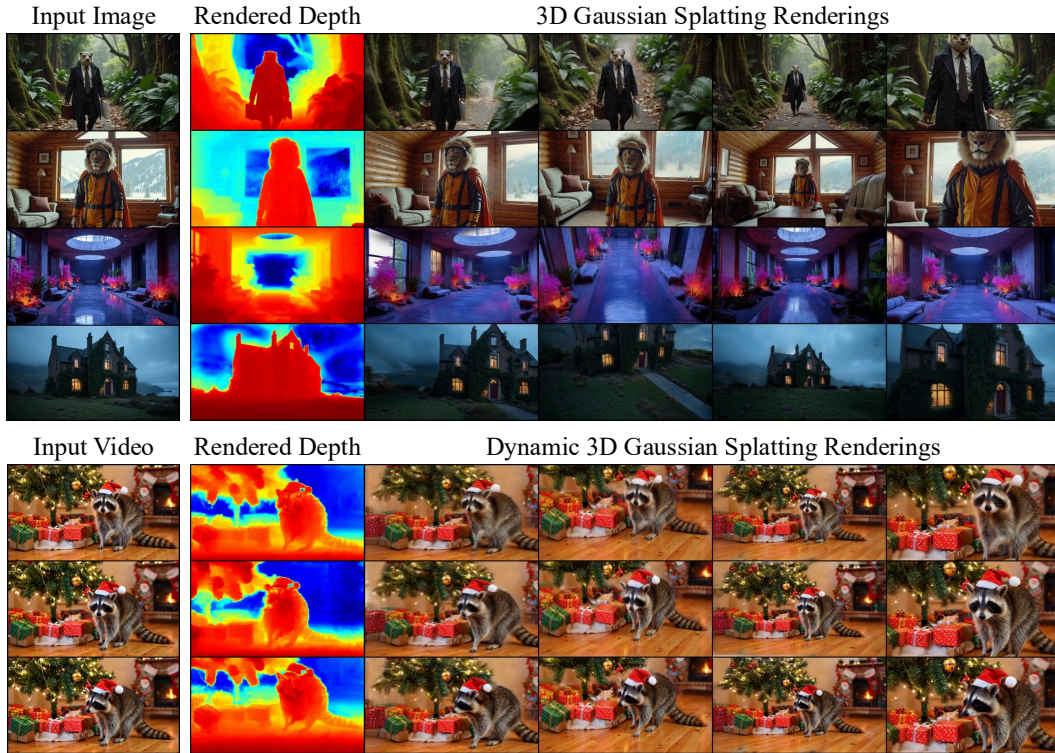


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in one feed-forward pass by self-distilling a video diffusion model without real-world multi-view data. From a video input (bottom), *Lyra* infers a dynamic 3DGS with interactive control over both time (rows) and viewpoint (columns).

ABSTRACT

Generating virtual environments is a central capability for gaming and for physical AI domains such as robotics, autonomous driving, and industrial AI. Current learning-based 3D reconstruction pipelines still depend on captured real-world multi-view data, a requirement that sharply limits their scalability and coverage. Video diffusion models now provide strong generative priors and broad scene imagination, but their 2D outputs prevent direct use in simulations where agents must navigate and interact with explicit environments. We present a self-distillation framework that distills the implicit 3D knowledge of video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, removing the need for multi-view training data. Concretely, we augment the standard RGB decoder with a 3DGS decoder and supervise this decoder directly from RGB outputs produced by the video model. This design trains the 3DGS decoder purely from synthetic data generated by video diffusion models while preserving an explicit, renderable scene representation. At inference time, our model synthesizes 3D scenes from either a text prompt or a single image and supports real-time rendering. The same framework extends naturally to dynamic 3D scene generation from a monocular

input video. Experiments demonstrate state-of-the-art performance for both static and dynamic 3D scene generation, establishing self-distillation from video diffusion models as an effective route to scalable generative 3D reconstruction.

1 INTRODUCTION

Creating high-quality 3D environments at scale remains a core challenge in computer vision and graphics, and it is increasingly important for film-making, VR/AR, and physical AI with closed-loop simulation. These applications require explicit 3D representations that render in real time, support physical interaction, and maintain consistent multi-view structure.

Recent advances in neural 3D reconstruction (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) recover such representations from posed images with impressive quality. Their dependence on accurate camera poses and high-quality image captures, however, fundamentally constrains scalability. Dynamic scenes further intensify this constraint because they often require synchronized multi-camera setups (Li et al., 2022). Reconstruction-only methods are also bound to observed content and cannot reliably extrapolate beyond the input views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) reduce per-scene optimization, but they inherit a different bottleneck: diverse, large-scale 3D training data remain scarce, which limits out-of-domain generalization.

Video diffusion models (Cosmos, 2025; Wan, 2025) offer a decisive way around this data bottleneck. Trained on massive internet-scale video corpora, they already capture high-fidelity appearance and broad scene variation. Because their training videos contain camera motion ranging from handheld capture to cinematic pans and drone footage, these models encode strong implicit cues about 3D world structure without explicit multi-view supervision. Unlike pure reconstruction approaches (Charatan et al., 2024), generative models can also synthesize plausible content outside the observed input. The missing piece is explicit geometry: video diffusion models output 2D frames, which prevents direct deployment in simulation and other tasks requiring geometric consistency, long-term coherence, and physical interaction. Recent camera-control results (Wang et al., 2024e; Ren et al., 2025) show that video models can be adapted to produce posed image sequences from relatively small posed datasets, turning them into a powerful source of both input and supervision for 3D reconstruction.

We use this capability to unify generation and reconstruction under *generative 3D scene reconstruction*. We introduce *Lyra*, a method that generates explicit 3D environments from the latent representations of video diffusion models in a single forward pass. Crucially, *Lyra* uses a self-distillation framework (Fig. 2) in which a 3D Gaussian Splatting (3DGS) decoder operates directly in the latent space of a video diffusion model. Given a single input image or video, we sample a camera trajectory, denoise the corresponding video latent, and decode it through two coordinated branches. The standard RGB decoder synthesizes the video sequence, while our 3DGS decoder produces an explicit 3D representation. The RGB branch acts as the teacher and the 3DGS branch as the student. To expand viewpoint coverage, we sample multiple camera trajectories, generate multiple video latents, and train the 3DGS decoder to fuse their information while reducing long-horizon and multi-view inconsistencies.

This generative reconstruction framework delivers three concrete advantages: **(i)** it produces large-scale synthetic environments across diverse scenarios directly from video diffusion models, eliminating dependence on real-world multi-view captures; **(ii)** it aggregates many views efficiently in the latent space of a video model, avoiding the memory burden of pixel-space feed-forward reconstruction; and **(iii)** it outputs explicit 3DGS scenes with geometric consistency, making the representations immediately useful for downstream settings such as physical simulation. Consequently, at inference time, *Lyra* generates high-quality 3D Gaussian scenes from monocular input that support real-time rendering, *without requiring any additional optimization or post-processing*.

We further extend the same self-distillation principle to dynamic 4D generation from monocular video. In this setting, the video model provides space-time supervision and the student learns time-conditioned 3DGS representations for novel-view synthesis of dynamic scenes.

Overall, our work makes the following contributions:

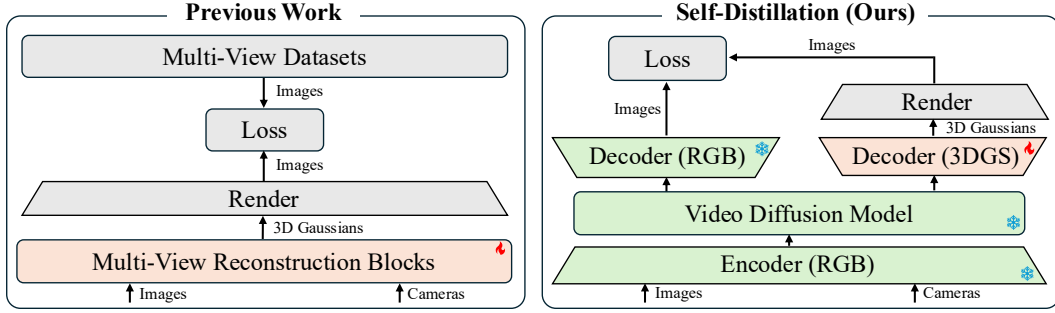


Figure 2: **Self-distillation framework of *Lyra***. Prior work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks on real-world datasets with limited diversity (Zhou et al., 2018; Ling et al., 2024b). Our self-distillation framework (right) instead trains generative 3D scene reconstruction from a pre-trained camera-controlled video diffusion model: the RGB decoder output acts as the teacher and supervises renderings from the 3DGS decoder student. A video model pre-trained on diverse 2D video data supplies broad multi-view supervision without real multi-view captures.

- We introduce a self-distillation framework that trains a 3DGS student model from camera-controlled video diffusion outputs, enabling explicit 3D generation without real-world multi-view training data.
- We design a latent-space 3DGS decoder that efficiently fuses hundreds of generated views and directly produces renderable Gaussian scenes in a single forward pass.
- We extend the framework to dynamic monocular videos, yielding a direct feed-forward route to 4D scene generation.
- We demonstrate state-of-the-art performance on static and dynamic 3D generation benchmarks, validating video diffusion self-distillation as a scalable training signal for generative reconstruction.

2 RELATED WORK

Multi-view image generation. Early multi-view image generation methods (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) primarily target object-centric scenes and often omit backgrounds. Subsequent work (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) scales multi-view generation toward full scenes. CAT3D (Gao et al., 2024b) adapts a pre-trained image diffusion model to generate multiple views from generated or real image inputs, then reconstructs those views into an explicit representation such as NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). Related space-time diffusion models extend this idea to dynamic 3D scenes from multiple viewpoints (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). These pipelines achieve strong visual results, but their explicit 3D grounding still depends on expensive per-scene optimization that is not amortized. Our work takes the more direct path: it generates the 3D scene itself from text or a single image instead of first generating multi-view images and then reconstructing them with optimization.

Camera-conditioned video generation. Camera-controlled video diffusion has advanced rapidly. MotionCtrl (Wang et al., 2024e) established camera conditioning by augmenting pre-trained video models with camera poses, and follow-up methods (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) strengthened this control with pixel-wise Plücker-coordinate conditioning. A parallel line explores training-free camera control for video diffusion models (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b). More recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) showed that large-scale synthetic training data can produce compelling camera-controlled video synthesis. Despite this progress, these methods still output videos rather than explicit 3D scenes. Our contribution is orthogonal and complementary: we convert camera-controlled video models into feed-forward 3D generators that support novel-view rendering *after* generation. Specifically, we build on GEN3C (Ren et al., 2025) and use it as the teacher in a self-distillation framework for explicit 3D reconstruction.

Feed-forward 3D models. Early feed-forward 3D reconstruction focused mainly on object-centric scenes without backgrounds, directly predicting NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) representations from text or images. Recent scene-level methods (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) regress per-pixel 3D Gaussians from one or more images, but they remain strongly

tied to the training distribution, such as RealEstate10K (Zhou et al., 2018), and generalize poorly to generated or out-of-distribution scenes. Newer generative reconstruction systems begin to close this gap: Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder for feed-forward reconstruction from generated multi-view images, and Wonderland (Liang et al., 2025a) uses a camera-controlled video model to produce 3D Gaussians with a feed-forward network. In contrast, our method uses the camera-controlled video model as a teacher and trains the student 3D model through self-distillation *without requiring real-world multi-view data*. We further show that the same framework extends directly to feed-forward 4D scene generation, a capability not addressed by prior work.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

Our central idea is to transfer the implicit 3D knowledge encoded in video diffusion models into an explicit 3DGS decoder that produces high-quality 3D representations. We realize this idea with a teacher–student framework: the video diffusion model acts as the teacher by generating RGB videos, and a 3DGS decoder operating in the same latent space learns as the student from those videos, as illustrated in Fig. 2. This section describes the video diffusion backbone (Sec. 3.1), the self-distillation procedure for training the 3DGS decoder (Sec. 3.2), the decoder design (Sec. 4), and the minimal extension that turns the pipeline into a dynamic 3D scene generator (Sec. 5).

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because our 3DGS decoder is supervised only by the camera-controlled video diffusion model, the teacher must provide strong 3D consistency. We therefore build on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model designed for 3D-consistent camera control. We summarize the GEN3C design choices that make it an effective teacher for our framework.

Spatiotemporal 3D cache. GEN3C improves video consistency and camera-control precision by constructing a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos. Each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimation (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is organized as an $L \times V$ array, where L denotes the temporal length and V denotes the number of camera views.

Rendering and structured guidance. GEN3C uses this cache by rendering each point cloud from arbitrary camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ denotes the rendering function that projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane according to the camera pose $\mathbf{C}^t$ at time t . Given a sequence of camera poses $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache elements produces V videos of length L , which are stacked into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks identify regions that the video diffusion model must complete, and the rendered cache supplies structured visual guidance for the generated video.

Video variational autoencoder (VAE). Diffusion-based video generators commonly use a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress videos into a lower-dimensional latent space for efficient training and inference. Given an RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, a pre-trained VAE encoder $\mathcal{E}$ encodes the video as $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$. Diffusion training and inference operate in this latent space, and the final video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is decoded with the pre-trained VAE decoder $\mathcal{D}_{rgb}$. We adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model. Its latent channel dimension is $C = 16$, its temporal dimension is $L' = (L - 1)/\tau + 1$, and its spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, with temporal compression factor $\tau = 8$ and spatial compression factor $\sigma = 8$.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as a student and use the camera-controlled video diffusion model $\mathcal{V}$ as the teacher. To obtain diverse supervision, we curate a large-scale prompt set with large language models (OpenAI, 2025; Bai et al., 2023), synthesize images I with an image diffusion model (Black Forest Labs, 2024), and expand those images into multi-view sequences with GEN3C (Ren et al., 2025).

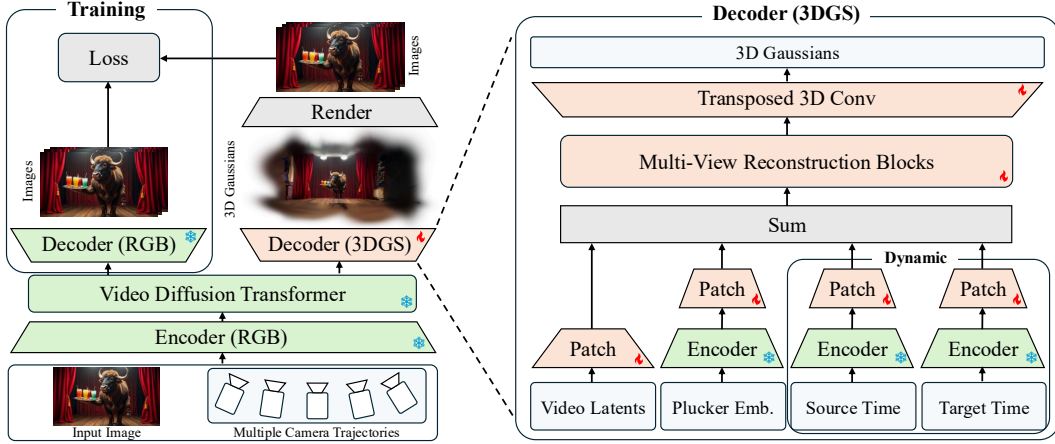


Figure 4: **3D Generative Reconstruction Framework.** The pipeline builds on a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large-scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning 2D renderings of generated 3DGS scenes with RGB-decoded generations from the frozen video model. Only the 3DGS decoder is trained; the pre-trained autoencoder and diffusion model remain fixed. At inference time, the 3DGS decoder directly produces the scene and no longer requires the RGB decoder. Time conditioning extends the same decoder design from static to dynamic 3D scene generation.

Teacher-student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ generates a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. We decode the latent $\mathbf{z}$ through two branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ produces video frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ outputs explicit 3D Gaussians $\mathbf{G}$. Student renderings are $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and are trained to match the teacher frames $\mathbf{I}_{\mathcal{D}_{rgb}}$, forming a direct self-distillation loop from generated video to explicit 3D.

Multi-trajectory supervision. To cover the scene beyond a single camera path, we sample $V = 6$ camera trajectories per input image, as shown in Fig. 3. For each trajectory v , we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ and $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ learns to fuse all $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$ while completing disoccluded regions. This design trains $\mathcal{D}_s$ entirely from synthetic supervision produced by $\mathcal{V}$; latent-space aggregation keeps the multi-trajectory problem tractable, and the explicit Gaussian output $\mathbf{G}$ enforces geometric consistency for simulation and real-time rendering.

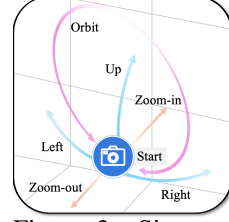


Figure 3: Six sampled camera trajectories provide broad view coverage for supervision.

4 FEED-FORWARD

RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

Our feed-forward 3DGS decoder $\mathcal{D}_s$ converts synthesized multi-view latents from the video model $\mathcal{V}$ into an explicit 3D representation that can be rendered from arbitrary viewpoints. We use 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) because it provides an explicit scene representation with fast rendering. Fig. 4 (right) shows the full decoder design.

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Existing feed-forward reconstruction frameworks generate 3D Gaussians per pixel, which directly limits the number and resolution of input views they can process. GS-LRM (Zhang et al., 2024e), for example, operates on 2–4 images at 512×512 resolution, and AnySplat (Jiang et al., 2025a) is trained with 24 images at 448×448 resolution. Our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution, far beyond the practical GPU-memory capacity of pixel-space methods. The limiting factor is attention over visual tokens, whose memory and compute scale with the number of pixels. We therefore avoid pixel-space scaling

and reconstruct directly in the compressed video latent space produced by the camera-controlled video diffusion model. The multi-view video latents are denoted as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ maps its inputs into the hidden dimension of the reconstruction blocks and outputs per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Its inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both represented in latent dimensions $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input component is patchified to match the hidden dimension; no additional visual encoder is required beyond a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) that converts the video latents into flattened reconstruction tokens. The summed inputs pass through reconstruction blocks following Long-LRM (Ziwen et al., 2024): each block contains one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers. We repeat the block twice, yielding 16 layers with 512 hidden dimensions. A transposed 3D convolution then maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is expressed as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. We compute raw Plücker embeddings from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. To align camera information with the video latents, we reuse the RGB encoder $\mathcal{E}$ from the pre-trained VAE. We separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins using $\mathcal{E}$, then concatenate the resulting latents along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of 2 so that the camera encoding matches the video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We train the 3DGS decoder with image-based reconstruction losses against the teacher RGB outputs. The reconstruction objective combines Mean Squared Error (MSE) loss $\mathcal{L}_{mse}$ with LPIPS loss $\mathcal{L}_{lpips}$ (Zhang et al., 2018) computed using VGG (Simonyan & Zisserman, 2014) features.

Depth supervision. RGB-only training can produce flattened geometry. We therefore add depth supervision using consistent video depth estimated by the off-the-shelf system ViPE (Huang et al., 2025) together with the scale-invariant depth loss $\mathcal{L}_{depth}$ from Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Following Long-LRM (Ziwen et al., 2024), we apply L1 regularization to opacity through $\mathcal{L}_{opacity}$ and remove the Gaussians with the lowest 80% opacity.

Total loss. The final objective is the weighted sum of all losses with weight factors λ_i

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where we set $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

The same framework extends to dynamic 3D scenes with only targeted architectural and data changes.

Self-distillation for dynamic scenes. The dynamic setup preserves the static design while replacing the single input image with a monocular video of length L and its camera poses. The video model then generates multi-view video latents that capture the same underlying motion from different viewpoints. For video inputs, we follow the static data protocol: sample diverse text prompts with large language models, generate videos using video diffusion models (Cosmos, 2025; Wan, 2025), and annotate the generated videos with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. We adopt the bullet-time design of Liang et al. (2025b) and define the dynamic 3DGS decoder $\mathcal{D}_s$ to generate time-dependent 3D Gaussians for all video frames. The decoder $\mathcal{D}_s$ follows $\mathcal{D}_s$ but augments the $\mathcal{D}_s$ input with encoded source and target time embeddings $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The raw source times are assigned to each input frame, forming $\mathbf{T}^{\text{src,raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, and the target time is $\mathbf{T}^{\text{tgt,raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both are

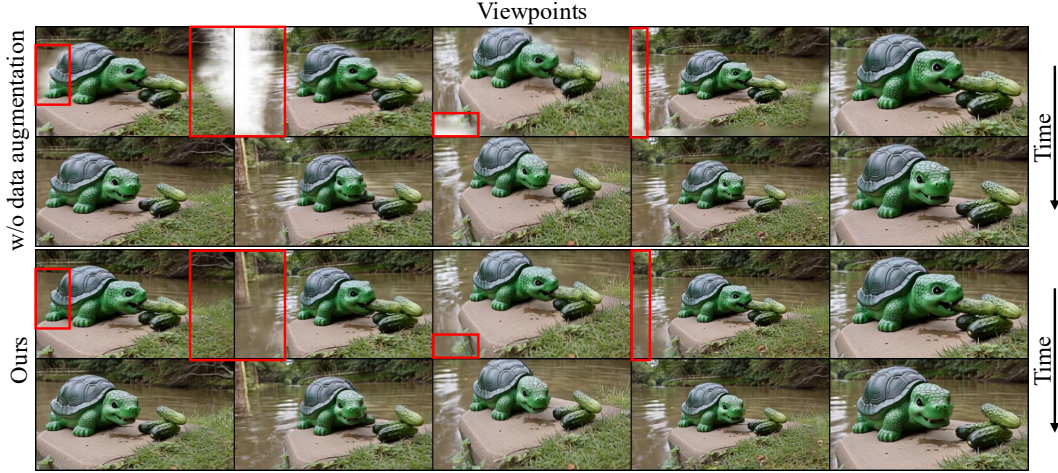


Figure 5: **Dynamic data augmentation.** Naive supervision of the time-aware 3DGS decoder produces artifacts in the generated 3D Gaussians: early timesteps from extreme poses have low opacity in regions outside the supervision signal. Our motion-reversed video augmentation gives each timestep full spatial coverage and prevents these early-timestep low-opacity artifacts.

spatially repeated, augmented with a 2-dimensional sinusoidal embedding to expand the channel dimension to 3, and encoded with the RGB encoder $\mathcal{E}$ to latent dimensions $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, we randomly sample a target timestep, decode the corresponding 3D Gaussians, and supervise them across all trajectories generated by dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from a pre-trained $\mathcal{D}_s$ and initialize the patchification layers of $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ to zeros. The time embeddings $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are added directly to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. Static scenes allow frames from different timesteps to supervise the same 3DGS rendering because the scene does not change over time. Dynamic scenes require stricter supervision: the 3DGS at a target time should only match frames from the corresponding timestamp. This constraint can otherwise produce a degenerate solution in which the 3DGS at one timestep ignores information from other timesteps (see Fig. 5).

We prevent this collapse with a dynamic data augmentation strategy that balances supervision across time. Early frames are naturally paired with viewpoints near the input image, whereas later frames correspond to more distant viewpoints; our augmentation provides each timestep with paired near and far supervision views. Concretely, we reverse the input video in frame order and pass it through the video model $\mathcal{V}$, producing six additional multi-view sequences. Reversing these sequences back to the original motion order yields six inward trajectories, moving from far viewpoints toward the input. Together with the original six outward trajectories, this produces 12 supervision views per timestep. The augmentation is used only during training: the reversed trajectories serve as additional supervision that prevents collapse from pruning artifacts and are not required at inference time.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. We train without any existing multi-view datasets. Instead, we construct the *Lyra* dataset and use only our video model to supervise the 3DGS decoder. The 3D reconstruction setting contains 59,031 images, and the 4D setting contains 7,378 videos. The data are generated from diverse text prompts spanning indoor and outdoor environments, humans, animals, realistic scenes, and imaginative content. For each image (3D) or video (4D), we synthesize six camera trajectories, producing 354,186 videos for 3D training and 44,268 videos for 4D training.

Baselines. We compare against state-of-the-art generative single-image-to-3D reconstruction methods: ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Because *no source code is available* for evaluating

Table 1: **State-of-the-art comparisons.** Our method is compared with prior single-image-to-3D generation systems on RealEstate10K, DL3DV, and Tanks-and-Temples.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336



Figure 6: **Image-to-3DGS Generation.** Five rendered views demonstrate the multi-view consistency of our generated 3DGS scenes.

these methods on our out-of-distribution evaluation set, we primarily use the quantitative comparisons reported in the corresponding papers.

Evaluation. We evaluate single-image-to-3D performance on RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017), following the protocols used by Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b). We report the standard reconstruction metrics PSNR, SSIM, and LPIPS.

6.2 MAIN RESULTS

Quantitative results. Tab. 1 reports single-image-to-3D quantitative results. Our method achieves the best performance on every benchmark and every metric, demonstrating that self-distillation from a camera-controlled video model produces a stronger feed-forward 3D generator. These results also indicate a clear scaling path: as video generative models improve in quality and consistency, our reconstruction framework can directly inherit those gains. Additional quantitative comparisons on the *Lyra* dataset are provided in Appendix C.

Qualitative results. Fig. 1 and Fig. 6 show novel-view renderings from generated 3DGS scenes; video results will be released upon acceptance. The renderings demonstrate that our method synthesizes high-quality unseen content for 3D/4D scenes while preserving consistency with the input image or video. Additional qualitative comparisons on the *Lyra* dataset appear in Appendix C.

6.3 ABLATIONS

Tab. 2 validates the main design choices on diverse out-of-distribution prompts. We compare 3DGS renderings against videos generated by our camera-controlled video diffusion model to isolate the effect of each component.

Real data only. Training only on real multi-view datasets, namely RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b), substantially reduces performance on out-of-distribution scenes.



Figure 7: **Ablations.** Rendering the same extreme novel viewpoint after image-to-3DGS generation makes the contribution of each component clear; additional depth visualizations and ablations appear in Appendix C.4.

This confirms that commonly used real multi-view datasets do not provide the diversity needed for our target generative setting.

Self-distillation + real data. Joint training with self-distillation and real data does not improve over self-distillation alone. The result shows that the synthetic supervision from the video model is already diverse and consistent enough to train the reconstruction model effectively.

No depth loss. Depth loss directly addresses flattened geometry and also slightly improves image-based metrics. We visualize the resulting depth maps in Appendix C.4.

No opacity pruning. Keeping all Gaussians instead of pruning by opacity yields a less compact representation and slightly lower visual quality. Opacity pruning improves efficiency as well: rendering at $H = 704$ and $W = 1280$ takes 18ms with pruning versus 30ms without pruning, a $1.67\times$ speed up.

No LPIPS loss. LPIPS loss improves robustness to input inconsistencies and preserves high-frequency details, making it an important complement to pixel-level supervision.

No multi-view fusion. A naive alternative generates 3D Gaussians for each camera trajectory independently and fuses them into a single point cloud. The ablation shows that learned multi-view fusion through reconstruction blocks is substantially stronger, because tokens from different trajectories can exchange information before Gaussian prediction.

No Mamba-2. Replacing our joint Transformer/Mamba-2 design with Transformer-only blocks slightly lowers quality and greatly increases runtime. One forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$ takes 3213ms with joint blocks compared with 20922ms with Transformer-only blocks, a $6.5\times$ speedup.

No latent-based 3DGS. Pixel-space approaches such as BTimer (Liang et al., 2025b) are limited to 12 input frames, whereas our latent-space decoder handles up to 726 input frames. Replacing latent-space 3DGS with a pixel-space decoder therefore runs out of memory, confirming that latent reconstruction is necessary for this scale.

7 CONCLUSION

We presented *Lyra*, a generative 3D and 4D reconstruction framework that produces explicit scenes from a single image or video. The key result is a self-distillation pipeline in which a camera-controlled video diffusion model teaches a latent-space 3DGS decoder, eliminating the need to collect real multi-view training data. The decoder directly reconstructs 3D Gaussians without per-scene optimization or post-processing, which enables scalable generation, broad scene coverage, and real-time rendering. Our dynamic extension shows that the same principle also supports monocular-video-to-4D generation.

Table 2: **Ablation study on the *Lyra* dataset showing the effect of each training and architecture component.**

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS		OOM	

The current scale and consistency of generated scenes remain bounded by the camera-controlled video diffusion teacher. This limitation is also a clear opportunity: stronger video diffusion models should translate directly into larger, more consistent 3D and 4D scenes within our framework. Future work can further expand scale by adapting auto-regressive techniques (Chen et al., 2024a), and motion quality may improve by explicitly modeling motion and tracking information inside the reconstruction network, as explored in concurrent work (Lin et al., 2025).

8 ETHICS STATEMENT

Like other generative AI technologies, our method could be misused to create misleading or synthetic 3D/4D content. We therefore frame this work around constructive applications: scalable simulation, robotics, and embodied AI data generation. In these domains, the method contributes concrete capabilities that are difficult to obtain with capture-based pipelines.

- **The ability to enable realistic simulation for embodied agents:** We provide controllable, physically consistent camera motion within generated 3D and 4D environments, supporting the training and evaluation of agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** Our method removes multi-view capture and expensive per-scene optimization, enabling high-fidelity, customizable scenes at scale for robotics, reinforcement learning, and closed-loop simulation research.
- **Better technical understanding:** This academic study provides a concrete case study of how camera control, spatiotemporal generative models, and explicit 3D reconstruction interact, informing future work in computer vision, graphics, and embodied AI.

We caution against deceptive or wrongful use of realistic generative outputs, particularly when misleading or unverifiable media could be produced. Safeguards such as provenance tracking, dataset documentation, and rigorous evaluation remain important for ensuring that generative 3D systems advance science and society responsibly.

9 REPRODUCIBILITY STATEMENT

To support full reproduction and accelerate follow-up research, we will release training and inference code, model weights, and data for both 3D and 4D generation upon acceptance.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section gives additional details on the GEN3C (Ren et al., 2025) video diffusion backbone introduced in Sec. 3.1 and on the refinement we use to strengthen its 3D consistency.

Conservative mask refinement. GEN3C’s forward warping can suffer from background leakage: occluded object parts in the source view (e.g., the lion body in Fig. 8) become visible from novel viewpoints but are not correctly marked in the disocclusion mask. This causes the video diffusion model to leave foreground regions incomplete, as shown in Fig. 8 (second and fourth columns).

We address this failure mode with a conservative masking strategy. Because the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unknown, we assume those areas may be occupied and derive refined disocclusion masks accordingly. This conservative assumption gives the video model the correct opportunity to complete potentially visible foreground regions that standard point-based rendering misses.

Our approach constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, similar to Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. The mesh acts as a boundary between visible and invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained via ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask from forward warping and ϵ is a small tolerance factor. When the mesh surface is closer than the original points, we mask out the corresponding pixels as potential foreground disocclusions. This produces more reliable guidance under large viewpoint changes, as shown in Fig. 8 (third and fifth columns).

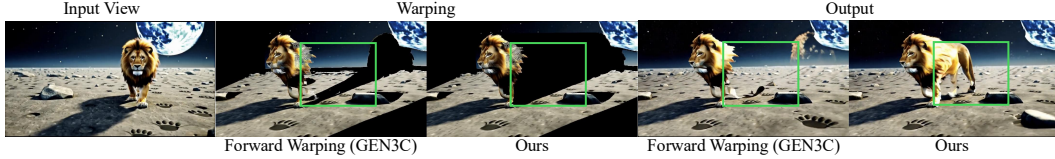


Figure 8: **Rendering function comparison.** Our model builds on GEN3C (Ren et al., 2025), which uses forward-warped images for camera-control conditioning. Forward warping (left) incorrectly fills the highlighted region with background pixels, whereas our conservative rendering function (right) masks it correctly and enables foreground completion.

B MODEL DETAILS

Progressive training. Training the 3DGS decoder directly on 6 trajectories of 121 frames at 704×1280 is expensive, so we use the progressive schedule in Tab. 3. We also initialize the dynamic model from the final static pre-trained model. Training takes 6 days on 8 NVIDIA A100 80 GB GPUs, and we use gsplat (Ye et al., 2025) for the 3DGS implementation.

Subsampling Gaussians. Pixel-aligned 3D Gaussian generation scales linearly with spatial and temporal resolution. Our generated scenes represent 726 frames at 704×1280 resolution, which would yield 654,213,120 per-pixel Gaussians. We therefore generate one 3D Gaussian per 8×8 spatial neighborhood, reducing the count by a factor of 64 to 10,222,080. Opacity-based pruning further reduces this number to 2,044,416, yielding a compact output representation.

Joint Transformer-Mamba blocks. We use the reconstruction block design from Long-LRM (Ziwen et al., 2024): one Transformer layer followed by seven Mamba-2 layers. Repeating the block twice

Table 3: **Progressive training setup used to scale from short, low-resolution inputs to full-resolution static and dynamic training.**

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1-6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

gives 16 layers with 512 hidden dimensions. Our network is substantially smaller than Long-LRM (Ziwen et al., 2024) or Wonderland (Liang et al., 2025a), which use 24 layers with 1024 hidden dimensions. While pure Transformers are often more expressive, they are much slower during training and inference than Mamba2. Under the same training budget, we find that Transformer-only blocks would likely require considerably more training iterations to match the rendering quality of the joint design.

Camera encoding. We encode cameras with Plücker embeddings, following prior work (Kant et al., 2024; Zhang et al., 2024e). The embeddings are computed at full spatial and temporal resolution and then projected into the video latent space with the pre-trained RGB encoder. Because the RGB encoder expects 3 input channels while Plücker embeddings are 6-dimensional, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins, then concatenate the resulting latents along the channel dimension.

Time encoding. For dynamic reconstruction, we make the 1-dimensional time input compatible with the RGB encoder by concatenating a 2-dimensional sinusoidal embedding to the original time value. We replicate these values spatially before encoding time into the video latent space. This construction lets the model encode source and target times independently in the same compressed representation used for visual latents.

Dynamic data augmentation. Fig. 9 visualizes our dynamic data augmentation for two example camera trajectories. The procedure creates paired supervision views that observe the same motion state from different extreme viewpoints, giving each timestep stronger multi-view supervision.

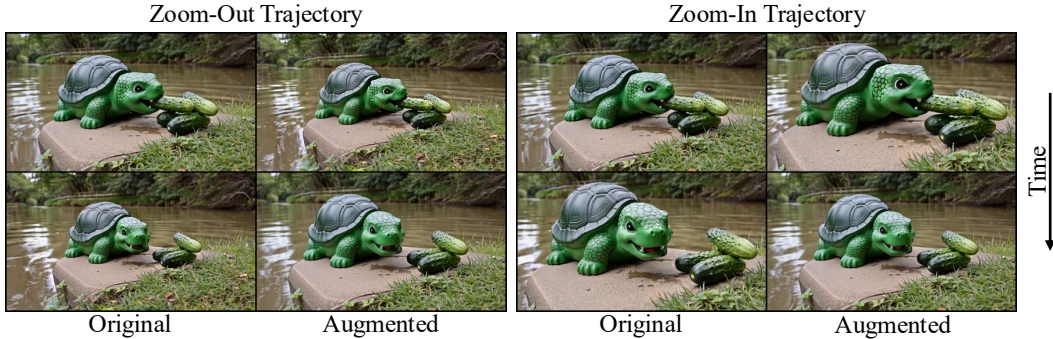


Figure 9: **Dynamic data augmentation videos.** Motion-reversed supervision gives each timestep full spatial coverage and prevents low-opacity artifacts in early timesteps. We show zoom-out and zoom-in trajectories together with their augmented videos, whose camera motion is flipped.



Figure 10: **Image-to-3DGS Comparison.** Five rendered views show that our method produces higher fidelity and substantially fewer artifacts than BTimer (GEN3C).

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare against BTimer (Liang et al., 2025b), a recent method for joint 3D and 4D reconstruction from posed images or videos. We contacted the BTimer authors to run experiments on the same scenes as *Lyra*. Because BTimer is purely regression-based, we integrate it with our GEN3C (Ren et al., 2025) video diffusion backbone: we use the same camera trajectories as our model and feed BTimer the RGB-decoded videos from GEN3C. BTimer operates in pixel space and requires 12 input images. As discussed in the main paper, pixel-space models such as BTimer run out of memory when using all 726 high-resolution GEN3C frames, so we uniformly subsample GEN3C outputs. For static evaluation, we select 12 frames that evenly cover the available viewpoints. For dynamic evaluation, we sample 11 frames spanning the viewpoint range and include the bullet-time frame corresponding to the target motion state. This guarantees that the reconstructed motion state is present as a reference. We denote this integrated baseline as BTimer (GEN3C), i.e., BTimer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 and Tab. 4a compare our method with BTimer (GEN3C). Our method produces higher-quality renderings with fewer artifacts.

C.3 DYNAMIC 3D EVALUATION

BTimer (GEN3C) comparison. For dynamic 3D evaluation, we compare with BTimer (GEN3C), described in Sec. C.1, on 100 out-of-distribution videos from the dynamic *Lyra* dataset. We crop videos to 512×512 for a fair comparison because BTimer was mainly trained at that resolution. Tab. 4b shows that our method substantially outperforms BTimer (GEN3C).

C.4 DEPTH VISUALIZATION

Fig. 11 visualizes depth from our generated 3DGS scenes. Depth supervision prevents flattened geometry while preserving visual quality.

Table 4: Quantitative results showing strong gains on both static and dynamic *Lyra* datasets.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) Static *Lyra* dataset comparison. (b) Dynamic *Lyra* dataset comparison.

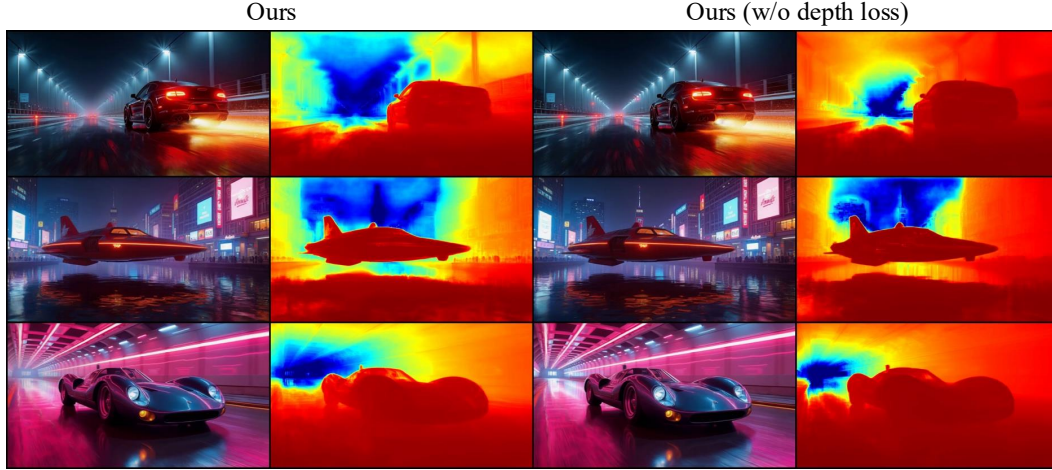


Figure 11: **Depth loss ablation.** The renderings and depth maps show that depth supervision prevents flattened geometry while preserving visual quality.

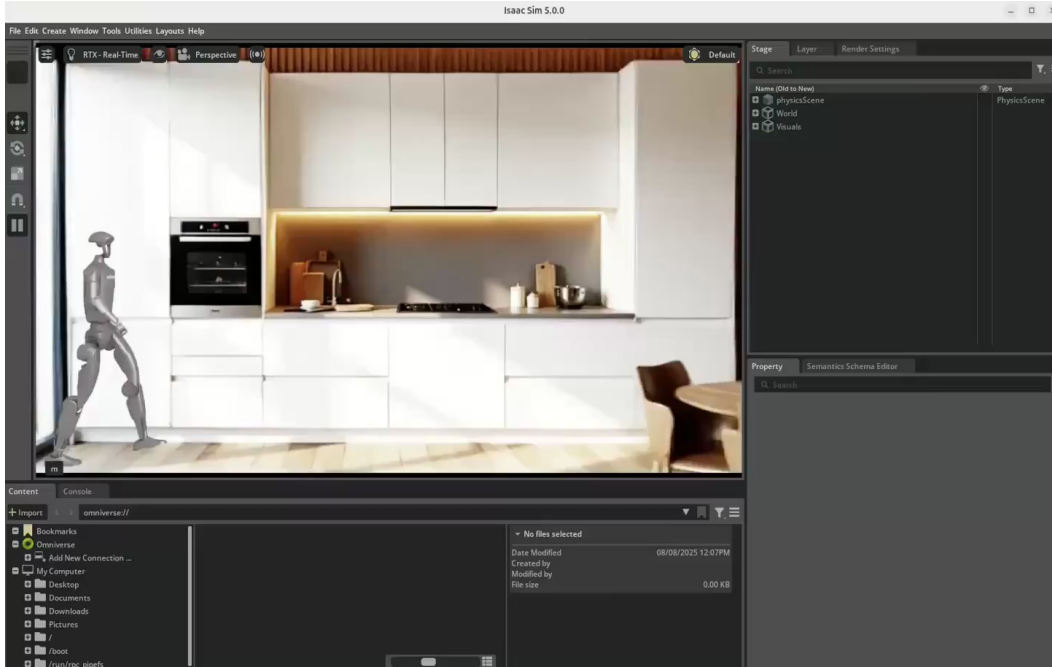


Figure 12: **Robot Simulation.** A generated 3DGS scene from our method can be imported into Isaac Sim 5.0 and used directly as a robot simulation environment.

D ROBOT SIMULATION

A primary motivation for our diverse 3D generator is scalable simulation for autonomous agents. We demonstrate an end-to-end pipeline: generate 3D scenes from text with our model, export the

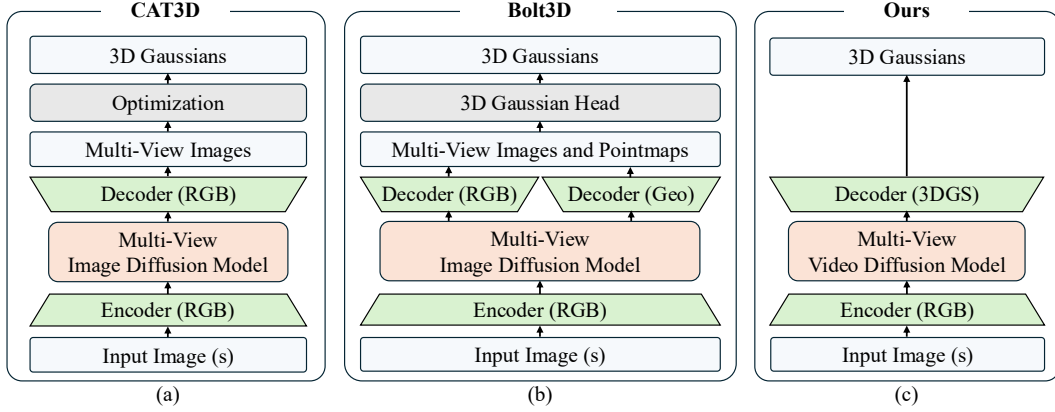


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) generates novel-view images and then reconstructs them into 3D Gaussians through optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps with a geometry autoencoder and predicts the 3D scene through a feed-forward 3D Gaussian head. Our approach takes a more direct route: it builds on a pre-trained camera-controlled video diffusion model and *directly* decodes multi-view video latents into 3D Gaussians.

3D Gaussians as a .ply file, and convert the .ply files into .usdz format. We use 3DGUT (Wu et al., 2025a) and its export function for .usdz conversion. The resulting .usdz file can be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment. These generated environments can support training and testing AI-based robots under diverse conditions. Early demo results will be released upon acceptance; Fig. 12 shows a screenshot of the interface.

E RELATED WORK

This section expands the related work discussion and positions our method against prior systems. Fig. 13 highlights the differences from CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b).

3D generation. Early 3D generation methods focused mainly on single object categories, extending GANs into 3D by using neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). CLIP-based supervision (Radford et al., 2021) expanded the field toward more flexible asset creation, including text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). Diffusion models then substantially improved quality by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

Several methods improve structural coherence by enforcing multi-view consistency during generation (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), while others use iterative inpainting for scene synthesis (Höllein et al., 2023; Shriram et al., 2024). A complementary direction lifts 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Feed-forward 3D methods directly predict 3D content from images or text inputs (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). In contrast, these approaches primarily produce static 3D scenes. Other specialized methods focus on particular scene classes such as faces (Kirschstein

et al., 2025). Concurrent methods (Liang et al., 2025b; Xu et al., 2025) address real scenes but do not handle diverse generated scenes or large viewpoint changes.

4D generation. 4D generation, or dynamic 3D scene synthesis, has progressed quickly. Many methods use text prompts or images as guidance; after the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent work improved both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

Beyond text conditioning, many methods lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Other efforts add physics priors (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), improve controllability with templates (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b), or pursue compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another line extends 3D GAN frameworks to 4D by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). These methods are generally limited by small-scale, single-category datasets, which restricts generalization. Most existing approaches are also object-centric and underrepresent backgrounds, so their overall photorealism remains behind recent video generation models. Our method directly leverages those stronger video priors.

Camera-conditioned video models. Camera control has become a major capability for video diffusion models. MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent methods (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) strengthen conditioning with Plücker coordinates, while others follow MotionCtrl and use flattened poses (Wang et al., 2025b). Training-free methods (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enable camera motion control without extra trainable parameters. Most of these systems rely on U-Net architectures, and other approaches explore low-rank camera-control adaptations (Zhang et al., 2025b).

Recent work continues to extend U-Net-based camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while diffusion-transformer approaches are emerging. Cheong et al. (Cheong et al., 2024) propose a transformer-based framework with limited scene and visual quality. DimensionX (Sun et al., 2024d) adds joint space-time control in diffusion transformers but uses pre-defined, non-continuous camera trajectories. Other work (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) studies pose and geometry estimation with video DiTs. CAT4D (Wu et al., 2025b) fine-tunes a multi-view backbone into a multi-view video diffusion model. Large-scale synthetic data further improves camera-controlled generation, as shown by SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a). 4DiM (Watson et al., 2024) instead trains a space-time diffusion model from scratch for single-image novel-view synthesis. Additional work retargets camera motion in videos with underlying motion (Seo et al., 2025; Ma et al., 2025). A prominent direction conditions camera-controlled generation with a spatiotemporal cache, such as point clouds or optical flow (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025); our framework turns this kind of video generator into an explicit 3D/4D scene generator.